
High Address
Memory Address Grows
Stack Grows Down
Low Address
Incoming Stack Arguments
Caller arguments passed on stack
Caller Frame
Stack Canary (Guard Slot)
Intended to guard saved frame records
Epilogue Checked
Local Variables & Spills
Static scalar & array local allocations
frame_pointer_rtx
Saved Return Address (LR')
Corrupted before canary is reached!
ATTACK TARGET
Saved FP' / SVE Registers
Frame pointer and vector registers
Callee-Saved
VLA Buffer (alloca / Dynamic)
Dynamic array allocated below saved LR'
OVERFLOW ORIGIN
Outgoing Stack Arguments
Space reserved for outgoing calls
Callee Args
SILENT DEFENSE FAILURE
- Overflow grows UP from VLA into saved LR'
- LR' corrupted & hijacked!
- Canary sits HIGHER above
Canary check PASSES!